Q3: The Zero Mode and the Proposed Horikawa Candidate

Continuation of the supplied October 8 research

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Abstract

We continue the search for a smooth closed non-Bach-flat absolute minimizer of $P(g) = \int |\operatorname{Ric}_g^\circ|^2$. The newly proposed Horikawa surface gives a valid fixed attainment problem, but its defining equation explicitly contains both (-3)-sphere and ADE vanishing-cycle topology. For the surviving $\mathcal{O}(-3)$ degeneration of the previous continuation, we solve the zero-mode radial inner problem exactly, including its finite renormalized quadratic minimum. We also compute the mixed bulk variation: an interior Euler–Lagrange term cancels an apparent point-jet shortcut. The inner minimum is negative, but the sign of the complete closed-manifold replacement is still undetermined. No smooth attainment theorem is asserted.

1 The fixed target and the new candidate

The target is a smooth absolute minimizer of P which is not Bach-flat. It suffices to prove attainment on a fixed manifold where all conformal classes are Yamabe-negative and no Einstein metric exists. Indeed a closed P-critical metric has constant scalar curvature; when this constant is nonzero, Bach-flatness and P-criticality together imply Einstein. These variational facts are inherited from the preceding work.

The user-provided proposal is

$$X_0 = \{w^2 + z^5 + x^{10} + y^{10} = 0\} \subset \mathbb{P}(1,1,2,5), \qquad M_H = X_0 \# \overline{\mathbb{CP}^2}.$$

The affine-cone gradient vanishes only at the origin; X_0 avoids the two singular coordinate points of the ambient weighted projective space. Weighted adjunction gives $K = \mathcal{O}_{X_0}(1)$ and $K^2 = 10/(1 \cdot 1 \cdot 2 \cdot 5) = 1$. The standard weighted hypersurface calculation gives $p_g = 2, q = 0$; this is the classical $(1,2)$ family [1]. Consequently

$$\chi(M_H) = 36, \qquad \tau(M_H) = -24, \qquad (b^+, b^-) = (5, 29), \qquad 2\chi + 3\tau = 0.$$

The ordinary sign classes $K \pm E$ provide a projection budget at least one for every metric. Combining the classical mixed monopole curvature bound [2] with the supplied strict negative-KE insertion gives

$$\frac{16\pi^2}{3} \le \inf_{g \in \operatorname{Met}(M_H)} P(g) < 8\pi^2. \tag{1}$$

The upper bound here is an inherited project result, not re-proved in this note. The positive actual monopole budget excludes nonnegative scalar-curvature representatives. Thus every conformal class is Yamabe-negative, and the lower bound excludes Einstein metrics. Horikawa therefore gives the same attainment reduction as the previously selected Godeaux blowup, with the advantage $b^+ > 1$ for ordinary Seiberg–Witten chamber bookkeeping.

Proposition 1 (Explicit topology in this particular Horikawa model)**.** *The surface X_0 contains an embedded smooth (-3)-sphere C with $K \cdot C = 1$. It also contains five disjoint A_4 Milnor-fibre regions; their vanishing-cycle lattices pair trivially with K. The blowup can be taken away from these configurations.*

Proof. Choose $c^5 = -1$ and $\epsilon \in \{1, -1\}$. The map

$$[u : v] \longmapsto [u : v : cv^2 : \epsilon i u^5]$$

respects the coordinate weights and lies in X_0, since $-u^{10} - v^{10} + u^{10} + v^{10} = 0$. Its first two coordinates recover $[u : v]$, so it embeds $\mathbb{CP}^1$. The curve lies in the weight-one charts, and $\mathcal{O}(1)$ restricts with degree one. Adjunction gives $-2 = C^2 + K \cdot C$, hence $C^2 = -3$.

For the second assertion, consider the proper family

$$X_a = \{w^2 + z^5 + x^{10} + y^{10} + ax^5y^5 = 0\}.$$

The singularity equations force $w = z = 0$, $x, y \neq 0$, and $2x^5 + ay^5 = 2y^5 + ax^5 = 0$. Thus the only singular parameters are $a = \pm 2$. At $a = 2$ there are five singular points, $w = z = 0$, $x^5 + y^5 = 0$. In local analytic coordinates their equation is

$$u^2 + v^2 + z^5 = 0,$$

the A_4 singularity. A nearby smooth fibre contains five disjoint Milnor fibres with the negative A_4 intersection matrix. Smooth fibres are diffeomorphic along paths in $\mathbb{C} \setminus \{-2, 2\}$, giving these regions in X_0. Their local canonical bundle is trivial, so K pairs to zero on their vanishing spheres. A blowup outside the chosen regions leaves all these pairings unchanged. □

The (-3) filling supports scalar-flat Kähler ALE metrics with $P = 8\pi^2/3$, while the A_4 topology supports hyperkähler ALE metrics with $P = 0$ [3, 4]. This proves that these topological possibilities are present; it does not prove that minimizing sequences develop them. Neither the interval (1) nor the displayed sign-class pairings alone excludes them. A change to Horikawa therefore does not supply the missing compactness theorem.

2 The remaining first-order zero mode

The previous continuation combined classical ALE mass flexibility with the supplied A02.59D boundary flux. For a marked $\mathcal{O}(-3)$ replacement at a smooth $\frac{1}{3}(1, 1)$ orbifold point of a P-critical bulk, it obtained

$$P(g_{\lambda,m}) = P(g) + \frac{8\pi^2}{3} - 24\pi^2 m\mu\lambda^2 + o(\lambda^2), \tag{2}$$

where m is Hein–LeBrun mass and $\mu = \langle (W^+ + sI/12)\zeta, \zeta \rangle$ on the invariant self-dual axis. Both mass signs are available on the same marked filling at the same complete P-energy. Under the stated marked local convergence and replacement hypotheses, minimality thus requires $\mu = 0$.

The scalar isotropy then gives

$$R^+ = W^+ + \frac{s}{12} I = \operatorname{diag}(\nu, \nu, 0), \qquad \nu = s/8.$$

The previously computed radial logarithmic term vanishes at this block. The next calculation treats its finite inner term. It uses the radial LeBrun filling, not the nonradial Stein fillings used for mass flexibility.

3 Exact solution of the radial inner problem

Normalize the $\mathcal{O}(-3)$ bolt radius to one and write

$$g_0 = \frac{dr^2}{G} + r^2(\sigma_1^2 + \sigma_2^2 + G\sigma_3^2), \qquad G = 1 + r^{-2} - 2r^{-4}.$$

The angular volume is $2\pi^2/3$. For

$$F = G + tu + O(t^2), \qquad H = G + tv + O(t^2), \qquad d = u - v,$$

use $g_t = dr^2/H + r^2(\sigma_1^2 + \sigma_2^2 + F\sigma_3^2)$. The previously derived radial Cartan formulas give the linearized self-dual curvature eigenvalues

$$L_1 = \frac{d'}{4r} + \frac{d}{2r^2}, \tag{3}$$

$$L_3 = -\frac{u''}{4} - \frac{7u'}{4r} - \frac{2u}{r^2} + L_1 + \frac{d'G'}{8G} + d\left(\frac{G'}{rG} + \frac{G''}{4G} - \frac{(G')^2}{8G^2}\right). \tag{4}$$

Smoothness at the fixed bolt requires $F(1) = H(1) = 0$ and $F'(1)H'(1) = 36$. At first order this reads

$$u(1) = v(1) = 0, \qquad u'(1) + v'(1) = 0.$$

Higher-order corrections can enforce the exact bolt product; they do not change the quadratic chiral energy on any fixed compact inner region, since $R^+(g_0) = 0$.

Proposition 2 (Radial linear response)**.** *Let $x = r^2$. The smooth-bolt linear problem $L_1 = \nu$, $L_3 = 0$ has the solution*

$$d_0 = \nu(x - x^{-1}), \tag{5}$$

$$u_0 = \nu\left[\frac{x}{6} - \frac{1}{2} + \frac{2 + \frac{1}{2}\log((x+2)/3)}{x} + \frac{-\frac{5}{3} - 2\log((x+2)/3)}{x^2}\right], \tag{6}$$

$$v_0 = u_0 - d_0.$$

It has the prescribed quadratic background $u_0 \sim \nu x/6$, $v_0 \sim -5\nu x/6$.

Proof. Equation (3) is $(d_x + d/x)/2 = \nu$. The bolt value fixes the integration constant and gives (5). Substitution in (4) reduces $L_3 = 0$ to

$$xu_{xx} + 4u_x + 2u/x = \nu\frac{2x^2 + 6x + 1}{2(x+2)^2}, \qquad u(1) = 0, \quad u_x(1) = \nu.$$

The substitution $u = y/x$ gives an elementary second-order equation; integration yields (6). Direct differentiation verifies $u_x(1) = \nu$, $v_x(1) = -\nu$ and the equation. At infinity,

$$u_0 = \nu\left[\frac{x}{6} - \frac{1}{2} + \frac{\frac{1}{2}\log x + 2 - \frac{1}{2}\log 3}{x} + \frac{-2\log x - \frac{2}{3} + 2\log 3}{x^2} + O(x^{-3})\right].$$

In particular the constant first correctors agree, as required by the vanishing logarithmic chiral-energy term. □

4 The exact finite inner minimum

Use the operator norm for Weyl curvature and put

$$\mathcal{C} = \int (4|W^+|^2 + s^2/12)\, dV = 4\int |R^+|^2\, dV.$$

On a closed manifold $P = \mathcal{C} - 8\pi^2(2\chi + 3\tau)$. For $X = R^2$, the coefficient of t^2 in the inner chiral integral is

$$\mathcal{Q}_X(u, d) = \frac{2\pi^2}{3}\int_1^X \left[x(d_x + d/x)^2 + 2xL_3^2\right] dx.$$

The corresponding background orbifold ball contributes $(2\pi^2/3)2\nu^2X^2$. Define the renormalized coefficient by subtracting this expression and then letting $X \to \infty$.

Proposition 3 (A minimum with fixed full asymptotic data)**.** *For smooth-bolt competitors for which $z = x(d - d_0)$ satisfies $z(1) = z(\infty) = 0$ and the following integrals converge,*

$$\mathcal{Q}_{\mathrm{ren}}(u,d) + \frac{4\pi^2}{3}\nu^2 = \frac{2\pi^2}{3}\int_1^\infty \left[\frac{(z')^2}{x} + 2xL_3(u,d)^2\right] dx \geq 0. \tag{7}$$

Equality holds for (5)–(6). Thus the exact minimum is

$$\boxed{\mathcal{Q}_{\mathrm{ren,min}} = -\frac{4\pi^2}{3}\nu^2 = -\frac{\pi^2 s^2}{48}.} \tag{8}$$

Proof. Since $d_x + d/x = 2\nu + z'/x$,

$$x(d_x + d/x)^2 = 4\nu^2 x + 4\nu z' + (z')^2/x.$$

Integration cancels the cross term by the two endpoint values of z. For the displayed solution, $L_3 = 0$ and $\mathcal{Q}_X = (2\pi^2/3)2\nu^2(X^2 - 1)$, proving the finite constant. The remaining homogeneous solutions for u are x^{-1} and x^{-2}; the zero bolt value and required derivative fix their coefficients. □

Boundary dependence. If $d = \nu x + c/x + o(x^{-1})$ instead, the right side of (7) acquires

$$\frac{2\pi^2}{3}\, 4\nu(c + \nu).$$

Thus the matching data at infinity are essential to the finite minimum. Equation (8) is an inner, successive-limit calculation: first differentiate at $t = 0$ on each fixed inner ball, then take its radius to infinity. For squared physical scale $\varepsilon = \lambda^2$ it corresponds to an inner contribution $-\pi^2 s^2 \varepsilon^2/48$. It is not yet the coefficient of a complete closed-manifold competitor. The outer response and the subleading matching terms have the same order. For example, simply cutting off a mass tail at radius r_0 allows an error of size $O(\varepsilon^2 r_0^{-4})$. The previously proved $o(\lambda^2)$ remainder in (2) does not control order λ^4.

5 The full mixed variation and its cancellation

One might try to change the bulk curvature jet slightly so that $\mu \neq 0$ and then apply (2). This loses bulk criticality. The missing term can be written explicitly.

Let E_g denote the variational gradient of $\mathcal{C}$, with $D\mathcal{C}_g[h] = \int \langle E_g, h\rangle$. Use the invariant mass tensor

$$B(x) = \frac{r^2 I - xx^{\mathsf{T}} - (J_0 x)(J_0 x)^{\mathsf{T}}}{r^4}, \qquad H_m = 9m\chi B,$$

with a fixed cutoff $\chi = 1$ near the puncture. The punctured first variation which gave (2) has, for a noncritical bulk,

$$a(g,m) = -24\pi^2 m\mu(g,p) + \int \langle E_g, H_m\rangle\, dV. \tag{9}$$

The latter integral is locally integrable. At a critical base metric, writing $L_g = DE_g$, its derivative is

$$\ell_m(h) = D_g a[h] = -24\pi^2 m D\mu_g[h] + \int \langle L_g h, H_m\rangle\, dV. \tag{10}$$

Contractions, volume and transported coordinate choices contribute no additional gradient term there, because $E_g = 0$.

Here is the consequence for a concentrated jet. Let $h_r(x) = r^2 h_1(x/r)$, with h_1 invariant, compactly supported and quadratic near zero. Its curvature variation $D\mu[h_r]$ can stay nonzero as $r \downarrow 0$. If $K_0 = D(R^+)_\delta$, then

$$D^2\mathcal{C}_g[h_r, h_r] = 8r^4 \int |K_0 h_1|^2 + O(r^6).$$

Nevertheless the complete coefficient satisfies

$$|\ell_m(h_r)| \le C|m|r^2. \tag{11}$$

Indeed $K_0 B = 0$ away from zero: B is the infinitesimal scalar-flat Kähler tail. In normal coordinates $g - \delta = O(|x|^2)$, so $D(R^+)_g[H_m] = O(|m||x|^{-2})$. Its product with $K_g h_r$ integrates to $O(|m|r^2)$. The other mixed terms are a bounded background curvature times $h_r \partial^2 H_m$, $H_m \partial^2 h_r$, or $\partial h_r \partial H_m$; they have the same integrable bound. This accounts for the moving norms and volume as well. Equivalently, integration by parts in the second term of (10) cancels the apparent point-evaluation term in the first.

For a nonzero principal quadratic form the resulting ratio is bounded:

$$\frac{|\ell_m(h_r)|^2}{D^2\mathcal{C}_g[h_r, h_r]} = O(m^2).$$

It does not diverge like r^{-4}. Thus a local point-jet variation cannot be used to overwhelm every finite gluing coefficient.

6 The actual remaining comparison

If a complete uniform expansion can be established at $\mu = 0$, it must have the form

$$P(G_{\lambda,t}) - P(g) - \frac{8\pi^2}{3} = \frac{1}{2}t^2 Q(h,h) + t\lambda^2 \ell_m(h) + \lambda^4 b_m + o(t^2 + |t|\lambda^2 + \lambda^4).$$

For $Q(h,h) > 0$, optimizing t then gives the finite coefficient

$$b_m - \frac{\ell_m(h)^2}{2Q(h,h)}.$$

The negative number in (8) supplies one computable inner part, under its specified matching conditions. It does not determine b_m, which must include the exterior Jacobi response, all boundary terms and the higher background jets. A nonnegative Hessian alone does not determine this total sign. No uniform fourth-order expansion or negative total coefficient has been established here.

For completeness, a return to purely explicit Kähler or homogeneous ansatzes does not give an immediate alternative. A closed Kähler P-critical metric has constant scalar curvature. When $s \neq 0$, varying the Kähler class makes $(c_1 \cdot \Omega)^2/\Omega^2$ stationary, forcing $c_1 \parallel \Omega$; harmonicity of the Ricci form then makes the metric Einstein. When $s = 0$ it is scalar-flat ASD. The complete homogeneous classification [5], Remark 1.5, likewise implies that a locally homogeneous $P = F_{-1/4}$ critical four-metric is Einstein or an opposite-curvature surface product. These candidates are all Bach-flat.

The fixed goal remains smooth global attainment. This continuation establishes explicit topology in the proposed alternative surface and an exact radial inner minimum; it also identifies the necessary mixed variation. The zero mode, other ALE topologies and general degeneration have not been excluded. No answer to Q3 is claimed.

References

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[5] M. Brozos-Vázquez, S. Caeiro-Oliveira, E. García-Río and R. Vázquez-Lorenzo, *Four-dimensional homogeneous critical metrics for quadratic curvature functionals*, arXiv:2309.02108.

[6] *Q3: Mass Variation and a Cyclic Degeneration Obstruction*, preceding October 8 continuation. Its first-order formula uses the supplied A02.59D boundary-flux calculation. The radial operators in (3)–(4) were derived in the preceding radial LeBrun analysis. This note independently differentiates the displayed solution and verifies the square-difference identity exactly.